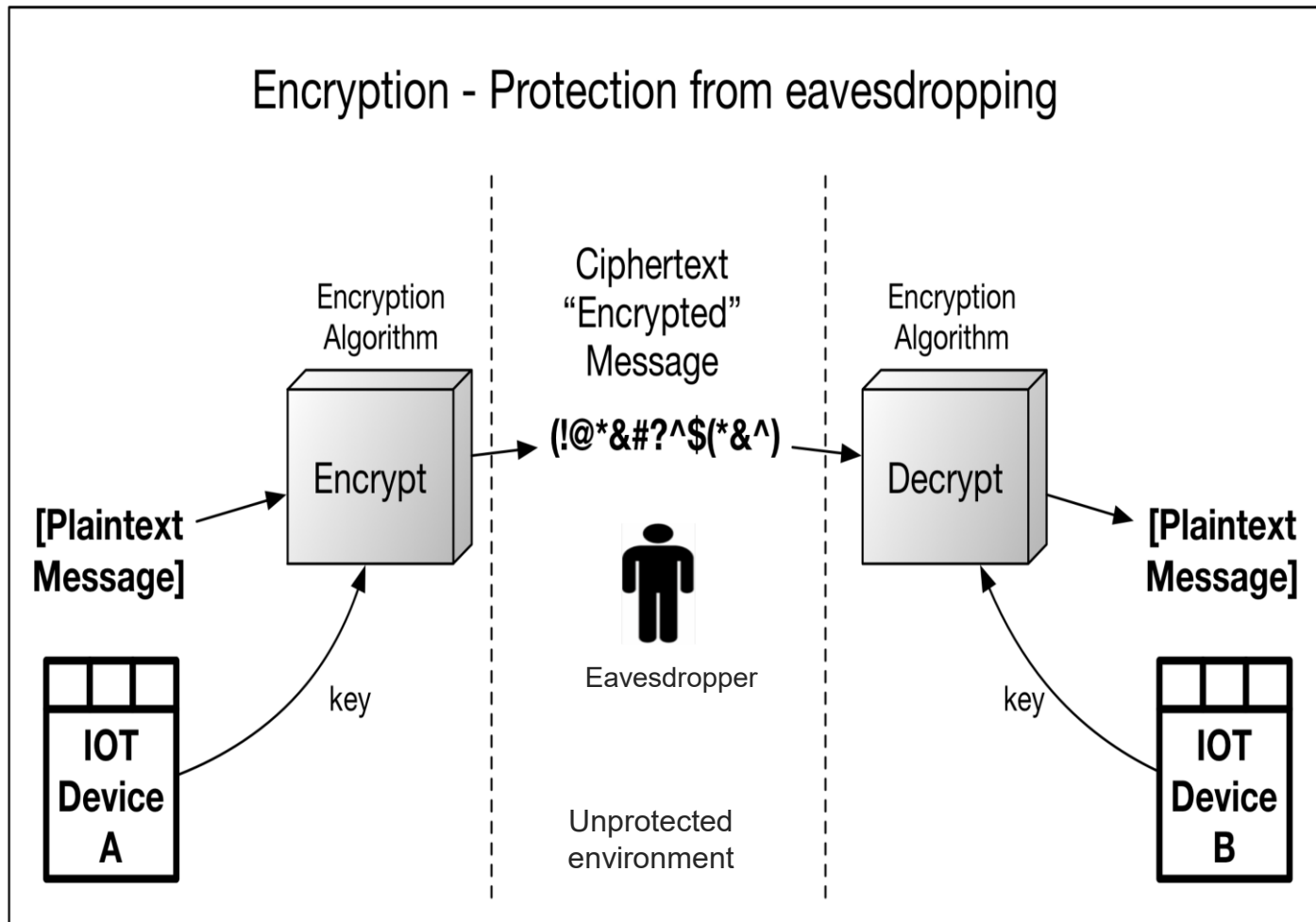
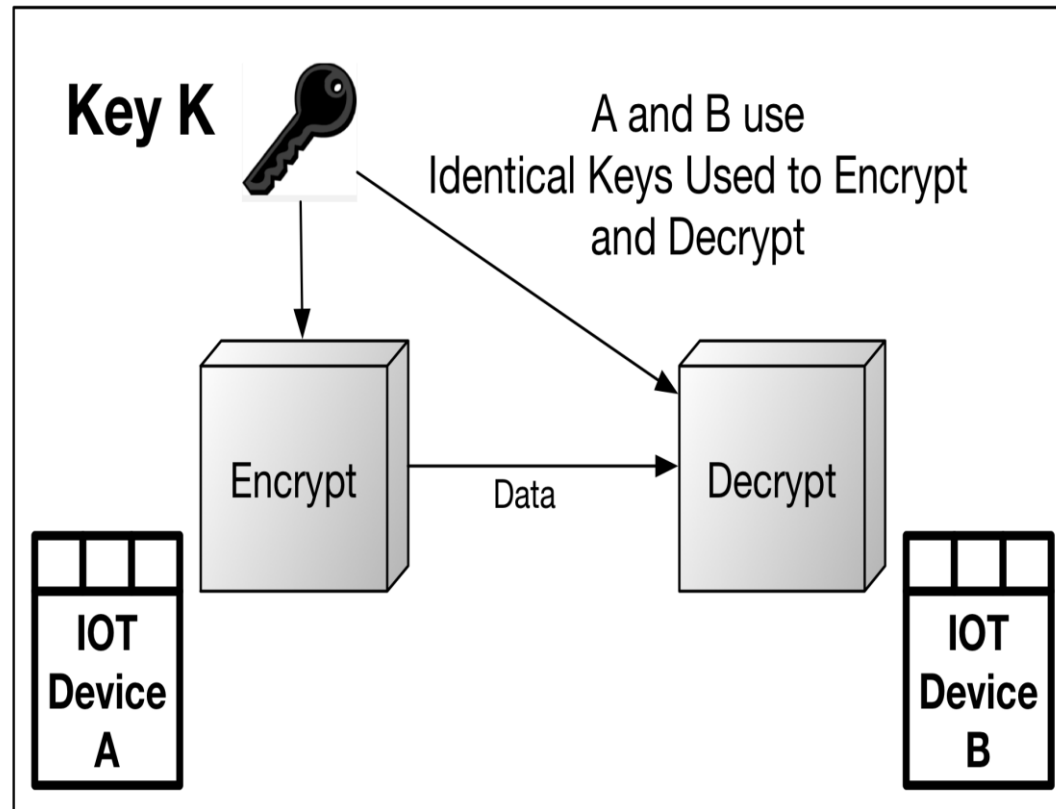


Security feature	Cryptographic service(s)
Confidentiality	Encryption
Integrity	Digital signature or Message Authentication Code (MAC)
Authentication	Digital signature or MAC
Non-repudiation	Digital signature

- Encryption (and decryption) (provides confidentiality):
 - Symmetric
 - Asymmetric
- Hashing
- Digital signatures (provide integrity, identity, and data-origin authentication as well as non-repudiation):
 - Symmetric: MAC used for integrity and data-origin authentication
 - Asymmetric: Elliptic Curve (EC) and Integer Factorization Cryptography (IFC)
- Random number generation (Most cryptography requires very large numbers originating from high-entropy sources. These are needed to generate or establish cryptographic keys and other tokens.)



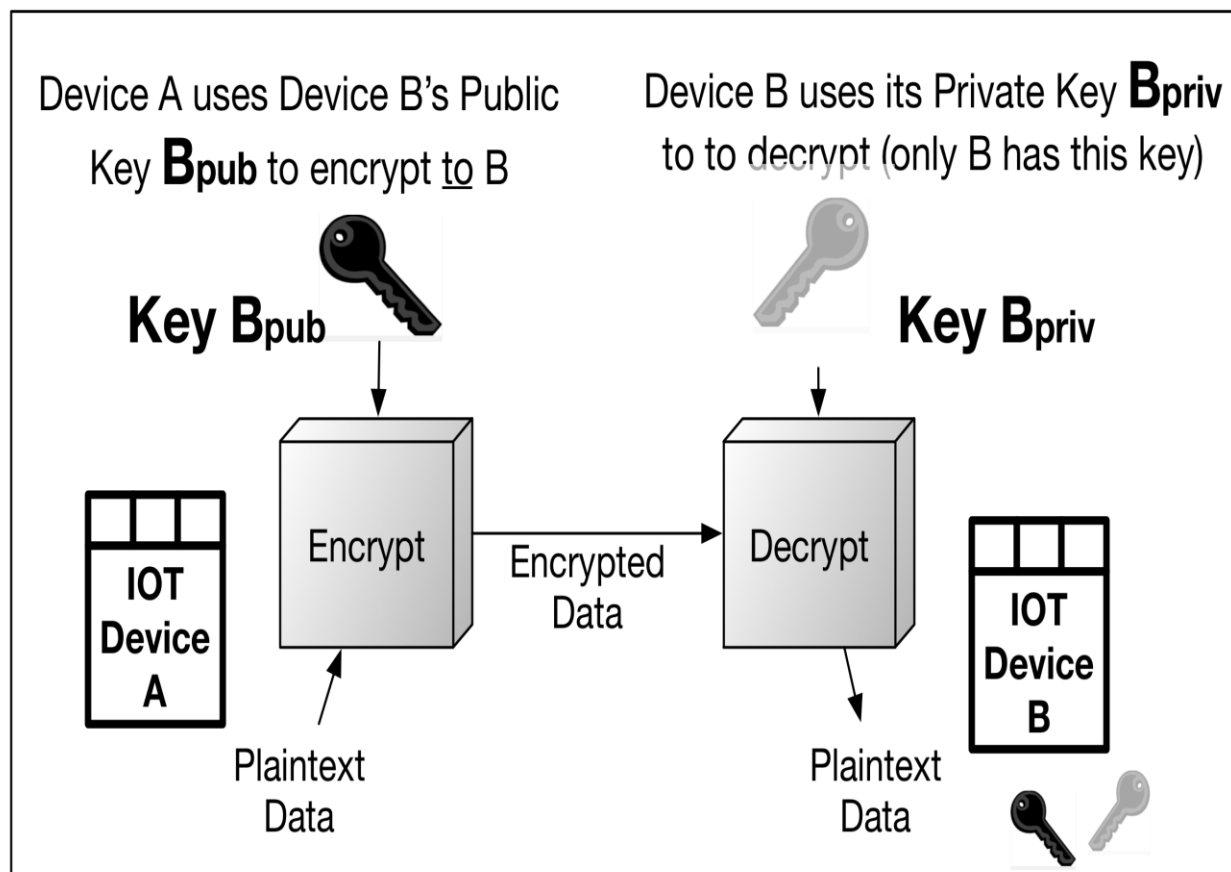
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Common symmetric ciphers include the following:

- Advanced Encryption Standard (AES) (based on Rijndael and specified in FIPS PUB 197)
- Blowfish
- DES and triple-DES
- Twofish
- CAST-128
- Camellia
- IDEA

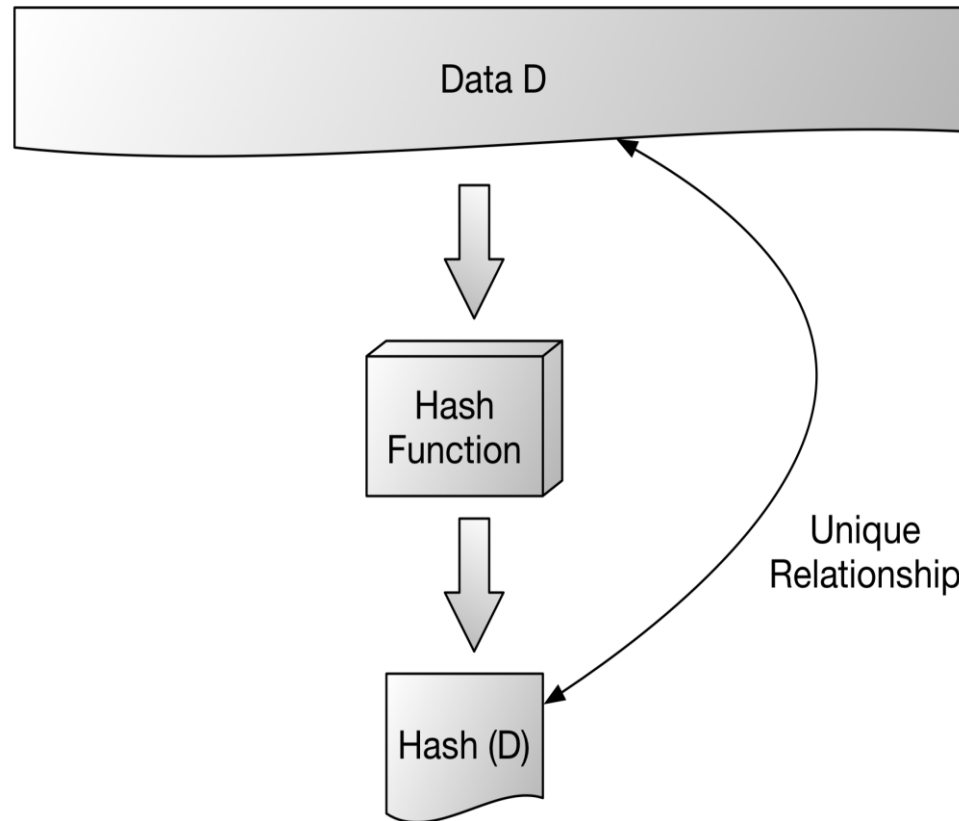
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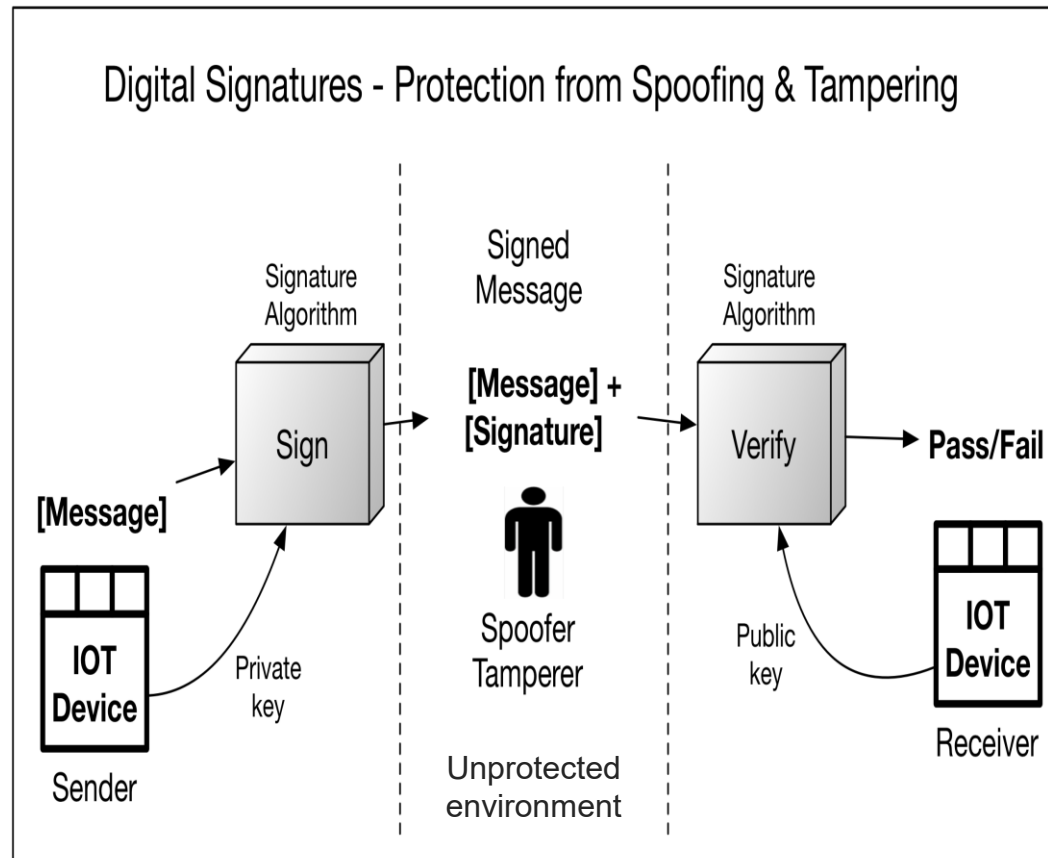
Common asymmetric encryption algorithms include the following:

- Rivest, Shamir, Adelman (RSA)
- Elliptic Curve Cryptography

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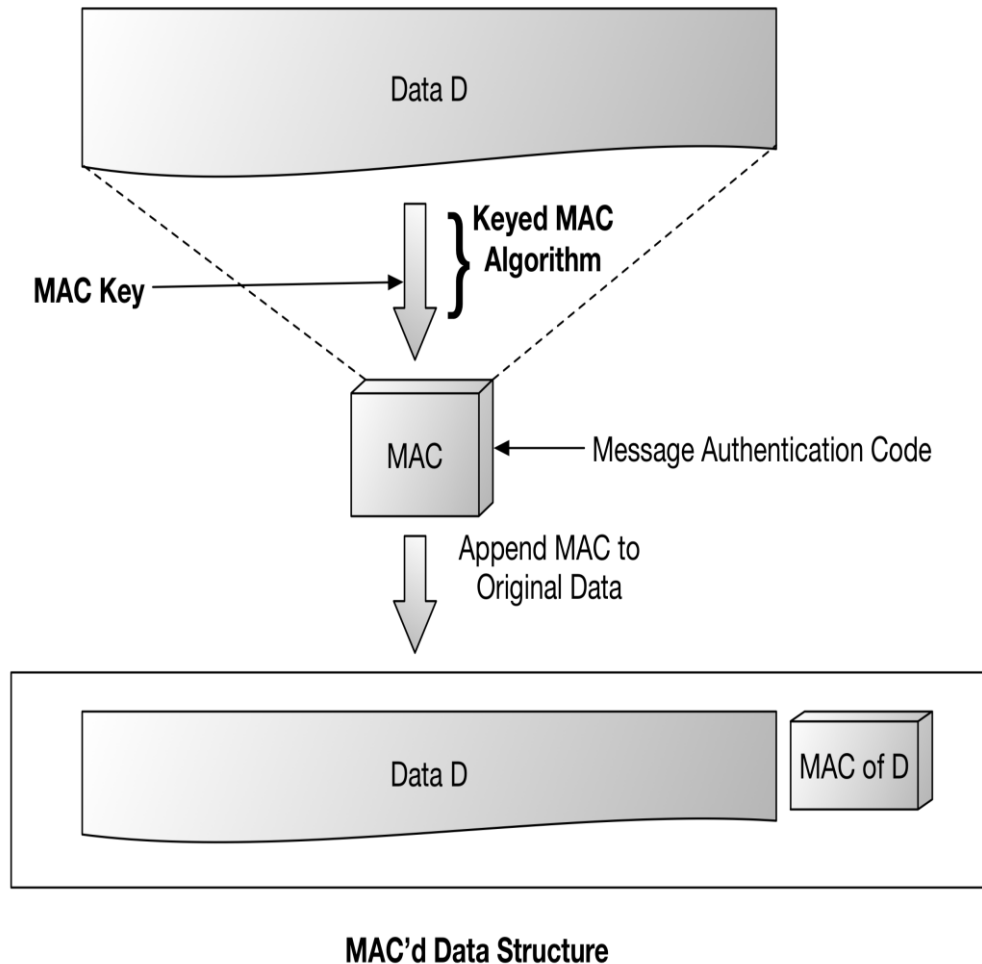


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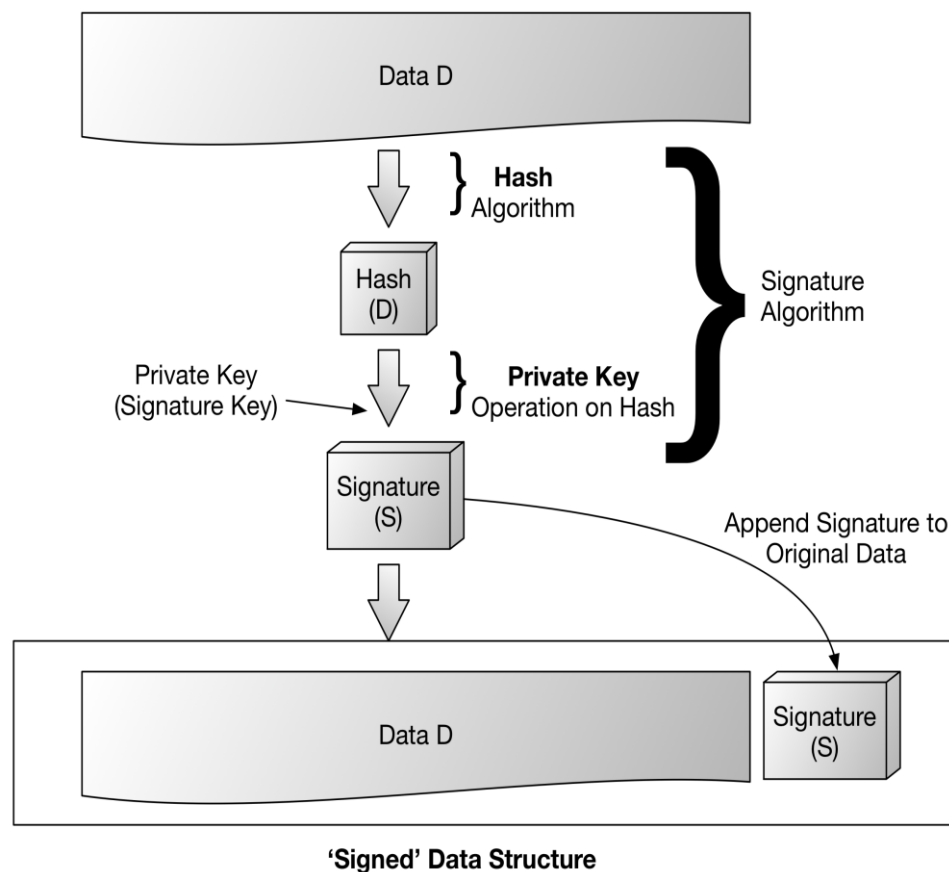
Symmetric Digital Signatures (MACs)



Examples of MACs include the following:

- HMAC-SHA1
- HMAC-SHA256
- CMAC (using a block cipher such as AES)
- Galois Message Authentication Code (GMAC) is the message authentication element of the GCM mode of AES)

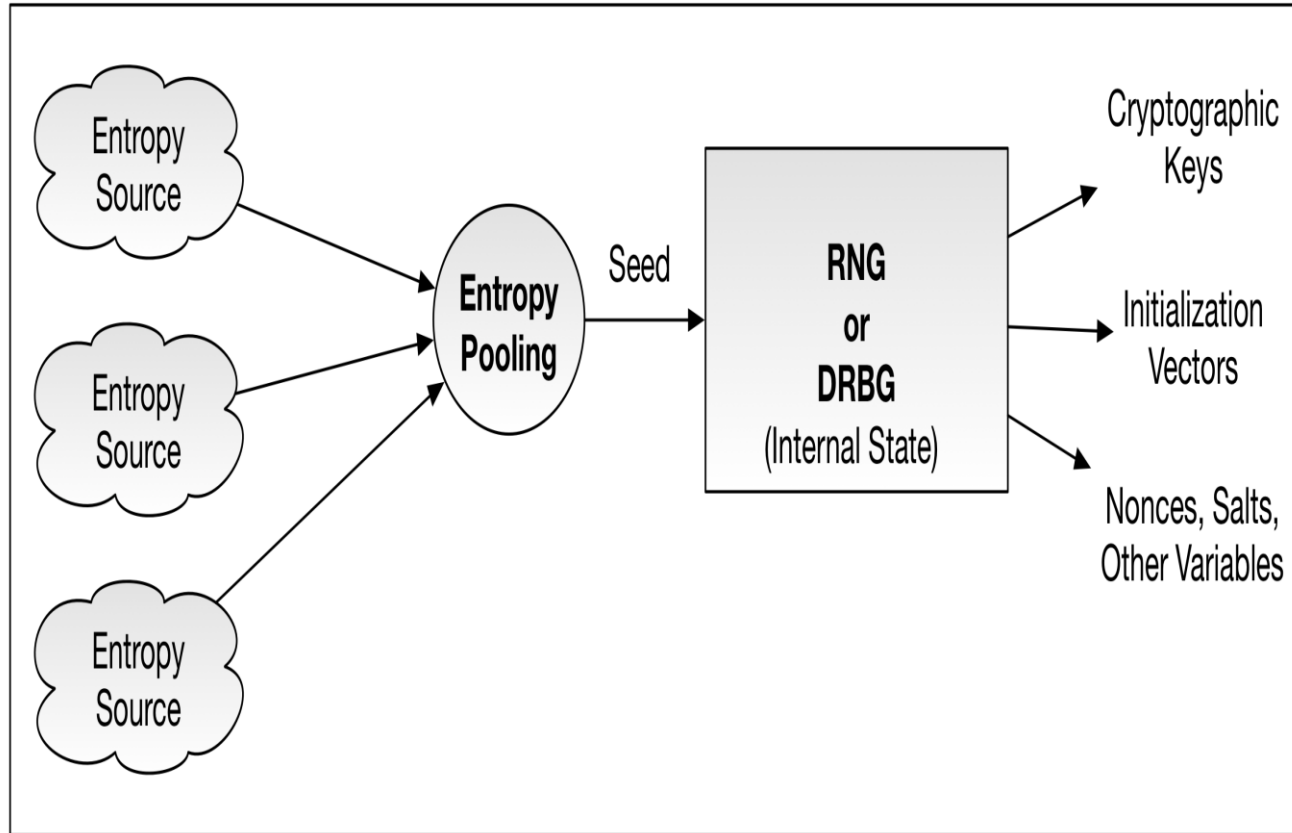
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Common asymmetric digital signature algorithms include the following:

- RSA (with various padding schemes such as PKCS1 or PSS)
- Digital Signature Algorithm (DSA)
- Elliptic Curve DSA (ECDSA)

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Deterministic Random-Bit
Generators (DRBGs)

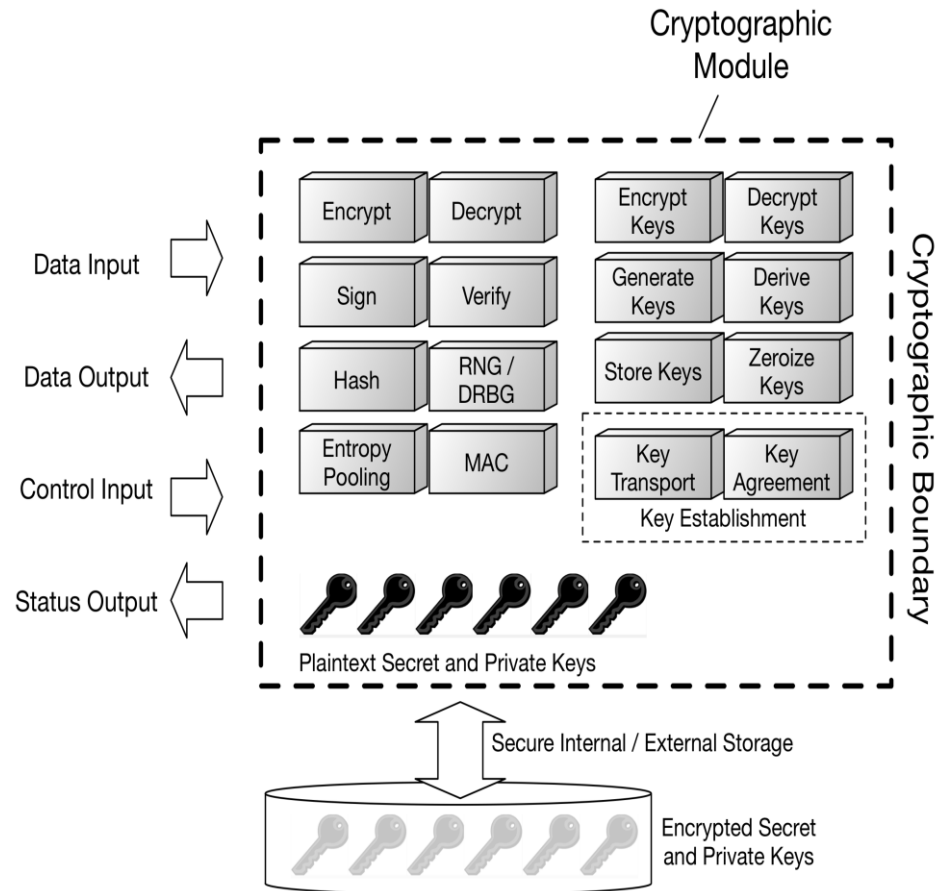
vs.

Non-Deterministic Random
Number Generators (NDRNGs)

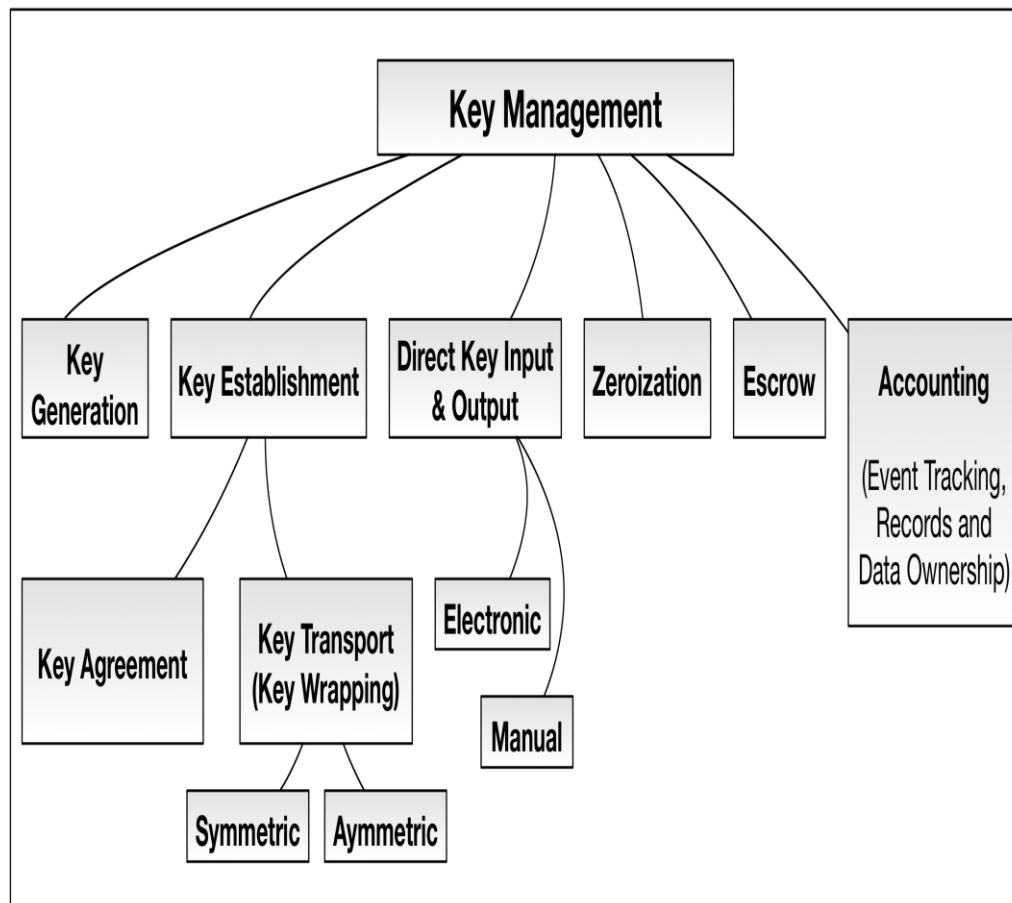
True Random-Number
Generators (TRNGs)

vs.

PseudoRandom-Number
Generators (PRNGs)



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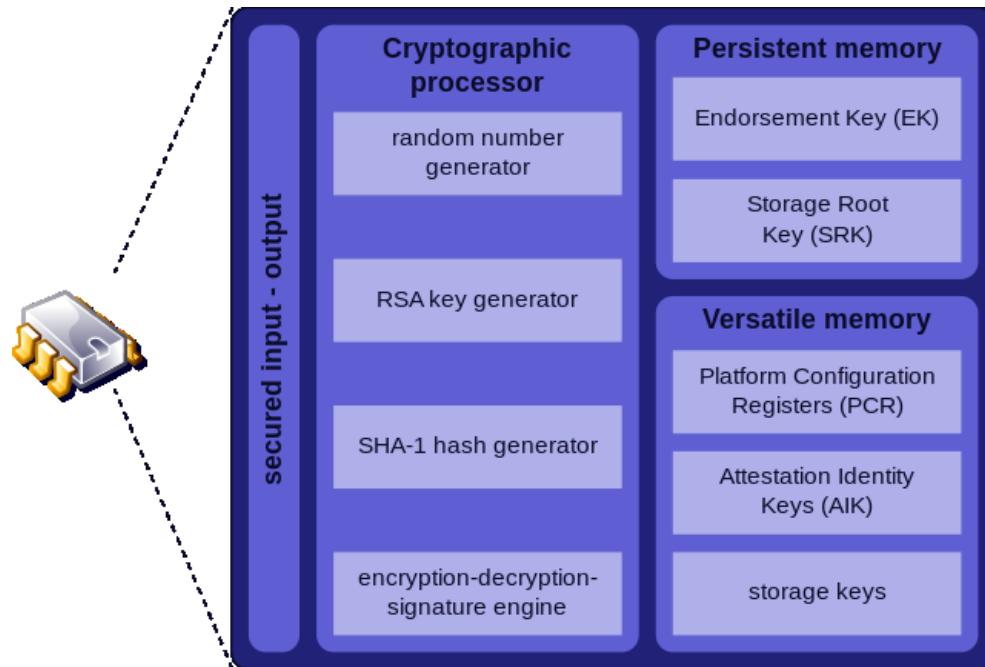
- Lightweight (hardware) security primitives
 - Memory-based Physical Unclonable Functions (PUFs)
 - True Random-Number Generators (TRNGs)
 - Trusted Platform Modules (TPMs), and other solutions
- Lightweight cryptography for the IoT: Standards and proposed solutions

- PUFs are real-world objects, such as hardware components, that have characteristics that allow for the identification of each instance of these components
- Thus, the relevant characteristics need to be **unique** per instance, and also persistent over time (**robust**)
- Ideally, these characteristics are also **hard to imitate** (copy) and hard to predict (**random**)
- Since memory forms an integral part of most systems, using a memory as a security primitive does not incur additional hardware costs
- Most often root privileges are required for the implementation of a memory-based Physical Unclonable Function

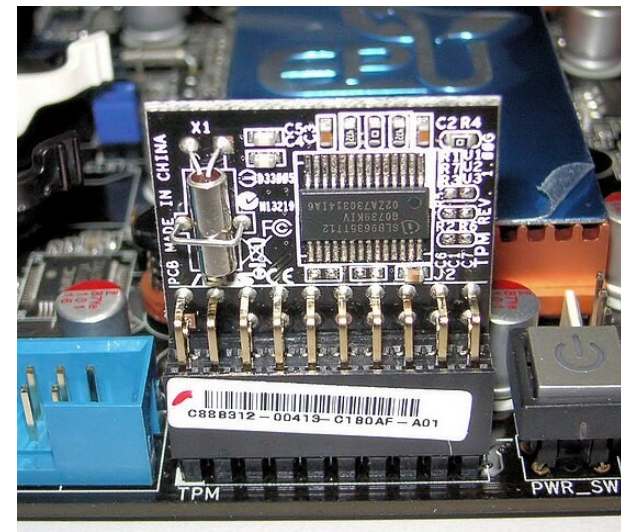
- Common examples of memory-based PUFs include:
 - The SRAM PUF that is based on the values of Static Random-Access Memory (SRAM) after a reboot.
 - The DRAM (Dynamic Random-Access Memory) PUF that is based on the same principle as the SRAM PUF (start-up values).
 - The DRAM PUF that is based on the retention/decay characteristics of the DRAM cells.
 - The DRAM PUF that is based on the data remanence characteristics of the DRAM cell.
 - The DRAM PUF that is based on the disturbances occurring in DRAM cells that are losing their charge due to nearby DRAM cells being constantly accessed (rowhammered).
 - The DRAM PUF that is based on the values attained from the DRAM cells when a reduced write or read latency time is used.
 - The Flash PUF that is based on the disturbances occurring in Flash memory cells that are not being accessed due to the constant programming of other nearby Flash memory cells.

- True Random Number Generators (TRNGs)
 - Usually based on physical processes, which may be microscopic or macroscopic
 - Common examples: Cards, coins, dice, roulette wheels, keyboard stroke timing, lava lamps, fishtank bubbles, radioactive decay, thermal noise, photoelectric effect
 - Can also be based on the random characteristics of memories, such as the writing or reading latencies, or the randomness of the initial values of memory cells
 - Since memory forms an integral part of most systems, using a memory as a security primitive does not incur additional hardware costs
 - Most often root privileges are required for the implementation of a memory-based TRNG
 - When TRNGs are based on quantum properties, such as the number (and other characteristics) of electrons that, for example, move between two components of a transistor, they are totally unpredictable
 - Unpredictable, but usually slow and potentially biased
 - Debiasing (post-processing) techniques may be required, such as von Neumann correction, parity-based and/or hash-based methods
 - In general, entropy distillation is used to convert biased random bits to unbiased random bits

- Trusted Platform Module (TPM)
 - An international standard for a secure cryptoprocessor
 - Dedicated microcontroller designed to secure hardware through integrated cryptographic keys
 - Usually requires the addition of hardware, which results in cost overheads



<https://en.wikipedia.org/wiki/File:TPM.svg>



https://en.wikipedia.org/wiki/File:TPM_Asus.jpg

- Trusted Platform Module (TPM)
 - Version 2.0
- General Solution: Dedicated hardware for cryptography and security
 - Incurs costs, in terms of money, area, power consumption, communication, etc.

TPM 2.0 - Features

- Standardized hardware/software (per TCG)
- Several chip vendors and all pin compatible
- RSA 2048-bit and ECC 256-bit supported
- Measured Boot supported with PCR
- Larger NV storage available (typically 50KB)
- Support for encrypted sessions for sensitive data over bus
- SPI interface (some vendors have an I2C option)
- Most vendors provide a mechanism for firmware updates
- Standardized endorsement key (EK) and certificate provisioning
- Advanced Policy management

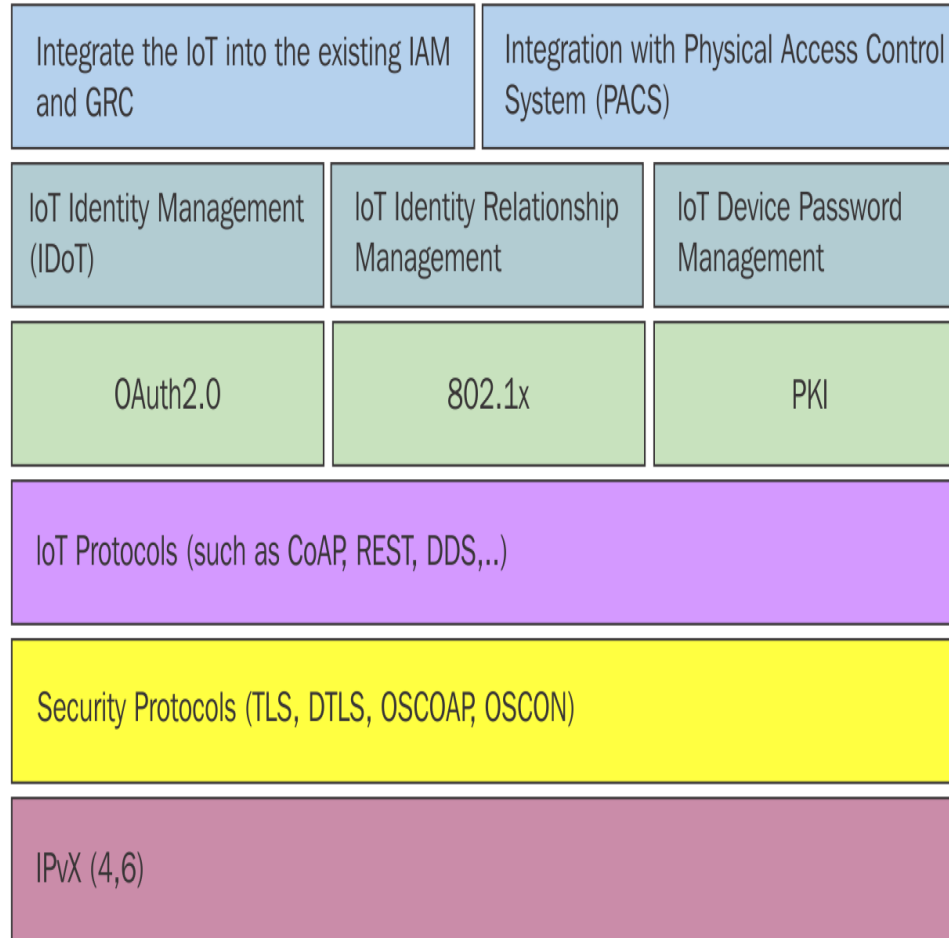
https://www.youtube.com/watch?v=2JbWuoK1r9M&ab_channel=wolfSSL

- Standards and proposed solutions
- Need for lightweight software solutions providing adequate security
 - Especially for resource-constrained devices (e.g., in the context of the IoT)
- Executive summary of NIST Internal Report on Lightweight Cryptography (<https://nvlpubs.nist.gov/nistpubs/ir/2017/NIST.IR.8114.pdf>):

“There are several emerging areas in which highly constrained devices are interconnected, working in concert to accomplish some task. Examples of these areas include: automotive systems, sensor networks, healthcare, distributed control systems, the **Internet of Things (IoT)**, cyber-physical systems, and the smart grid. Security and privacy can be very important in all of these areas. Because the majority of modern cryptographic algorithms were designed for desktop/server environments, many of these algorithms cannot be implemented in the constrained devices used by these applications. When current NIST-approved algorithms can be engineered to fit into the limited resources of constrained environments, their performance may not be acceptable. For these reasons, NIST started a lightweight cryptography project to investigate the issues and then develop a strategy for the standardization of lightweight cryptographic algorithms.”

- U.S. National Institute of Standards and Technology (NIST) lightweight cryptography standardisation process (2015 to 2023)
(<https://csrc.nist.gov/projects/lightweight-cryptography>)
 - Finalists announced in 2021 (<https://csrc.nist.gov/Projects/lightweight-cryptography/finalists>)
 - ASCON (<https://ascon.iaik.tugraz.at/>) family of algorithms was selected (February 2023)

- ASCON features:
 - Authenticated encryption and hashing (fixed or variable output length) with a single lightweight permutation
 - Provably secure mode with keyed finalization for additional robustness
 - Easy to implement in software and hardware
 - Lightweight for constrained devices: small state, simple permutation, robust mode
 - Fast in hardware
 - Fast in software: Pipelinable, bit-sliced 5-bit S-box for 64-bit architectures
 - Scalable for more conservative security or higher throughput
 - Timing resistance: No table look-ups or additions
 - Side-channel resistance: S-box optimized for countermeasures
 - Key size = tag size = security level (128 bits recommended)
 - Minimal overhead (ciphertext length = plaintext length)
 - Single-pass, online (encryption and decryption), nonce-based, inverse-free



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